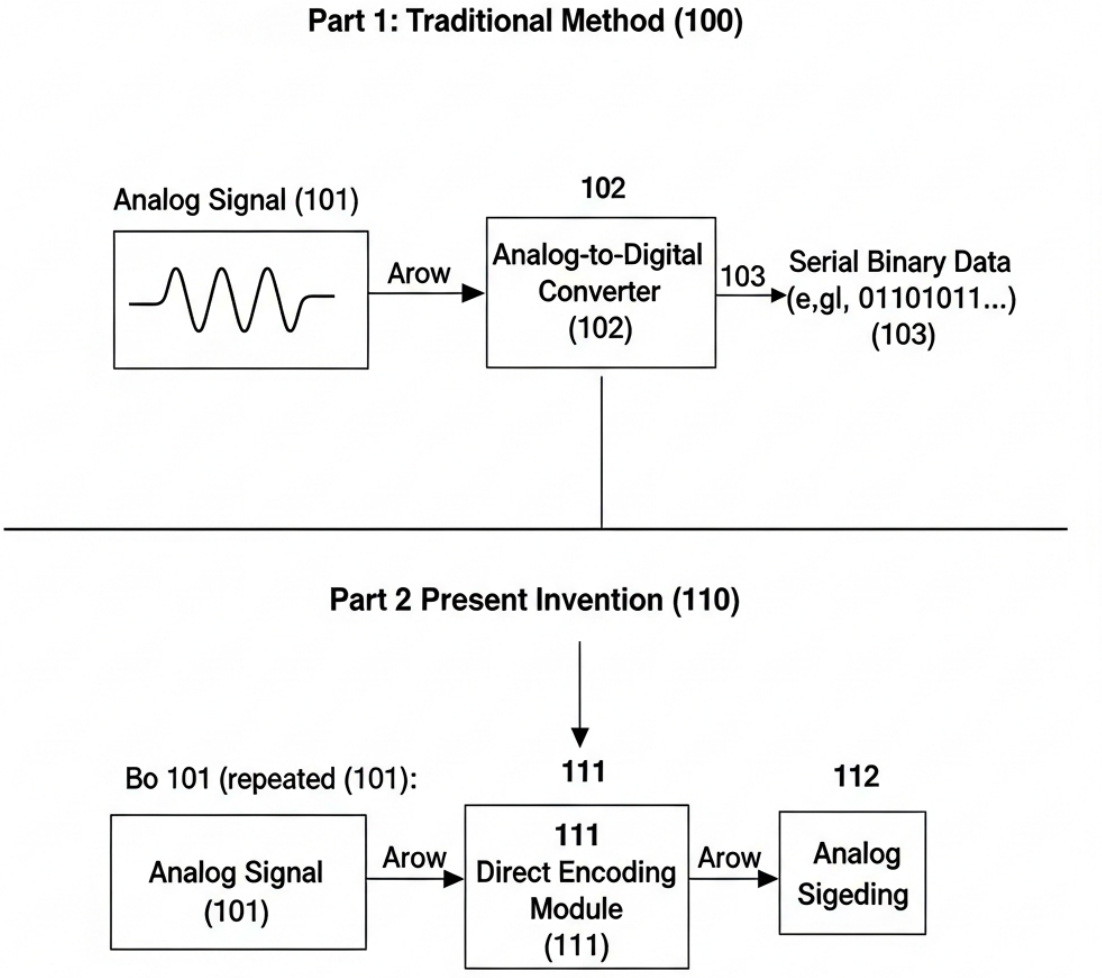

Part 1: Traditional Method (100)
Analog Signal (101)
102
Arow
Analog-to-Digital Converter (102)
103
Serial Binary Data (e,gl, 01101011...) (103)
Part 2 Present Invention (110)
Bo 101 (repeated (101):
111
112
Analog Signal (101)
Arow
111 Direct Encoding Module (111)
Arow
Analog Sigeding

FIG. 1

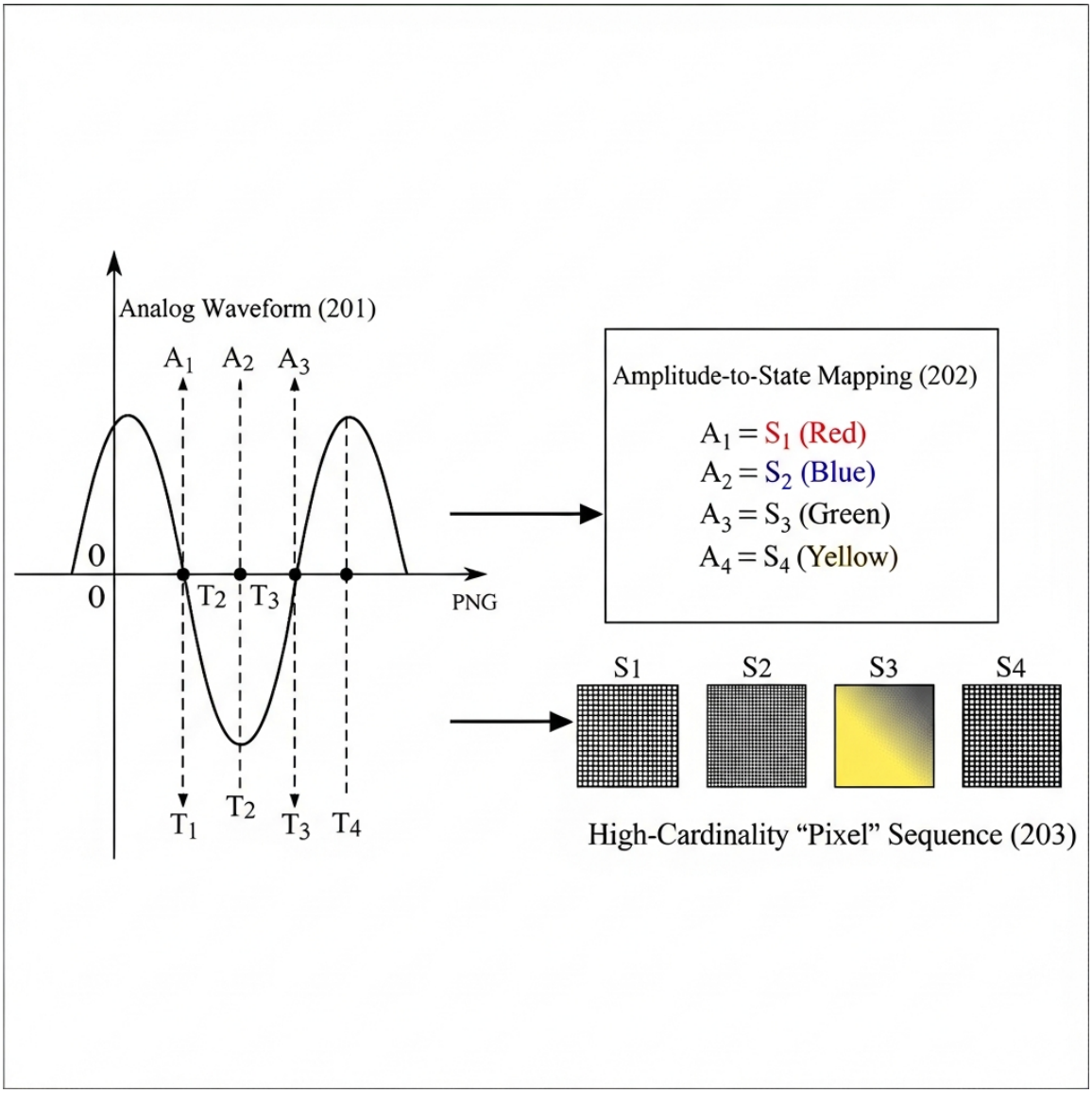

Analog Waveform (201)
A1
A2
A3
0
0
T2
T3
PNG
T1
T2
T3
T4
Amplitude-to-State Mapping (202)
A1 = S1 (Red)
A2 = S2 (Blue)
A3 = S3 (Green)
A4 = S4 (Yellow)
S1
S2
S3
S4
High-Cardinality "Pixel" Sequence (203)

FIG. 2